

PlantDoc - Plant Care Tracker

Comprehensive Project Architecture, Technical Documentation & Deployment Report

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1. Executive Summary & Project Purpose

PlantDoc (Plant Care Tracker) is an intelligent, responsive web platform designed to streamline houseplant care, watering schedules, environmental monitoring, and collection management for plant enthusiasts and administrative supervisors. The system combines real-time weather data integrations, dynamic care frequency calculation algorithms, streak tracking, location tagging, and comprehensive data visualization.

Key Deployment Endpoints:

- Live Frontend Web Application: <https://plant-watering-tracker-2026.web.app>
- Cloud Run REST Backend Service: <https://plant-care-service-358974981913.asia-south1.run.app>

2. Technology Stack & Architecture

Frontend SPA	React 19, Vite 8, React Router DOM 7	Single Page Application architecture with fast HMR
Icons & Charts	Lucide React, Recharts 3.10	Modern botanical iconography and interactive data visualizers
Styling	Vanilla CSS Custom Tokens, Responsive Grid/Flex	Pixel-perfect CSS tokens, glassmorphism cards, soft shadows
Backend API	Java Spring Boot, Google Cloud Run	RESTful backend handling users, plants, and activity historv
Hosting	Firebase Hosting	Global CDN edge distribution for static assets and web app

3. User Modules & Feature Overview

3.1 Interactive Dashboard (/dashboard)
Features 4 top summary metric cards (Total Plants, Need Watering, Overdue, Best Streak), Live Weather Widget (OpenWeather API integration with city selector and base water recommendation math), floating 3-column filter toolbar (Search for a plant..., All Plants location filter, All Status filter), covered white container box for MY PLANTS with bottom carousel controls, and right sidebar panels for UPCOMING WATERING and RECENT ACTIVITY.

3.2 My Plants Catalog (/my-plants)
Provides quick filter tabs (All, Need Watering, Overdue, Best Streak), floating white pill search and status inputs, 3-column responsive plant grid with rich plant cards featuring environmental metrics (Temperature, Humidity, Sunlight), care schedule box, care streak badge, and 6 action buttons (Watered Today, Upload Photo, Add Note, Edit Plant, View Details, Delete).

3.3 Care History & Analytics (/history, /analytics)
Real-time timeline feed tracking watering events, note additions, and plant creation logs with user attribution badges (by <userName>). Interactive Recharts graphs visualizing 7-day watering trends, status distribution percentages, and growth timelines.

3.4 User Settings & Security (/settings, /change-password)

Personal profile management (Display Name, Email), plant care defaults (location, watering cycle, reminder preferred time), dynamic Celsius (°C) and Fahrenheit (°F) temperature unit conversion, appearance theme switcher (Nature green, High contrast green, Dark botanical, System default), browser desktop push notification toggle, one-click CSV record exporter, local cache clearing, and dedicated Change Password page.

4. Administrative Management Panel (/admin)

The Admin Panel provides complete system oversight for super-users and administrators. Protected by role-based route guards (<AdminRoute>), the admin interface includes dedicated sub-modules:

4.1 Admin Dashboard (/admin)

Top profile header with circle settings button, admin profile pill, and exit logout button. 5 top metric summary cards (Total Users, Total Plants, Safe, Needs Water, Overdue). 7-day watering activity bar chart, plant health percentage progress meters, dashed Plants Needing Attention container, recent activity feed, and user/plant growth line chart.

4.2 Admin Plants Table (/admin/plants)

Floating filter bar with status dropdown, owner dropdown, location dropdown, and + Add Plant button. Covered white table container displaying plant photos, truncated UUID tags (ID: 0873bbf7-...), species, owner, location pin, watering frequency, last watered, next watering date, status pill, streak, and pagination footer.

4.3 Admin Users Table (/admin/users)

Search bar, role selector (User/Admin), status selector (Active/Inactive), and + Add User modal. Covered white table container showing user initials avatar circle, name, ID tag (ID: USR001), email, role badge, plant count, joined date, active status pill, action buttons (View details, Edit user), and pagination footer.

5. Data Algorithms & Business Logic

- Water Status Classification: Calculates days passed since lastWatered vs frequency. If days > frequency => 'Overdue'; if days >= frequency - 1 => 'Water Soon'; else => 'Safe'.
- Streak Calculation: Increments currentStreak upon consecutive on-time watering actions and updates bestStreak accordingly.
- Temperature Unit Conversion: Converts Celsius ranges (e.g. 18-27°C) to Fahrenheit (64-81°F) dynamically using regex pattern replacement: $F = (C * 9/5) + 32$.
- CSV Record Exporter: Generates RFC 4180 compliant CSV backup files containing plant attributes, frequencies, streaks, and timestamps for user download.
- Weather-Based Care Math: Adjusts base water dosage (mL) dynamically based on live temperature, humidity, rain probability, wind speed, and outdoor plant placement.

6. Production Build & Deployment Verification

The application was compiled using Vite production bundler (Vite 8.2) resulting in clean 0-error build outputs. Production bundle deployed to Firebase Hosting CDN edge nodes.

Summary Table:

Metric / Test Target	Status	Details
Vite Production Build	PASSED	Compiled in 1.10s with zero lint or runtime errors
Firebase Hosting Deploy	PASSED	Deployed live to plant-watering-tracker-2026.web.app
Responsive UI & Layouts	PASSED	100% pixel-perfect match to reference designs across mobile & desktop
Route Security & Guards	PASSED	Protected routes enforce authenticated session & role checks

